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— ONE —

The Basics

I have yet to see any problem, however complicated, which, when looked at in the right way, did not become still more complicated.

—POUL ANDERSON[1]

More Than the Sum of Its Parts

A system isn't just any old collection of things. A **system*** is an interconnected set of elements that is coherently organized in a way that achieves something. If you look at that definition closely for a minute, you can see that a system must consist of three kinds of things: *elements*, *interconnections*, and a *function* or *purpose*.

For example, the elements of your digestive system include teeth, enzymes, stomach, and intestines. They are interrelated through the physical flow of food, and through an elegant set of regulating chemical signals. The function of this system is to break down food into its basic nutrients and to transfer those nutrients into the bloodstream (another system), while discarding unusable wastes.

A football team is a system with elements such as players, coach, field, and ball. Its interconnections are the rules of the game, the coach's strategy, the players' communications, and the laws of physics that govern the motions of ball and players. The purpose of the team is to win games, or have fun, or get exercise, or make millions of dollars, or all of the above.

A school is a system. So is a city, and a factory, and a corporation, and a national economy. An animal is a system. A tree is a system, and a forest is a larger system that encompasses subsystems of trees and animals. The earth is a system. So is the solar system; so is a galaxy. Systems can be embedded in systems, which are embedded in yet other systems.

Is there anything that is not a system? Yes—a conglomeration without any particular interconnections or function. Sand scattered on a road by happenstance is not, itself, a system. You can add sand or take away sand and you still have just sand on the road. Arbitrarily add or take away football players, or pieces of your digestive system, and you quickly no longer have the same system.

When a living creature dies, it loses its "system-ness." The multiple interrelations that held it together no longer function, and it dissipates, although its material remains part of a larger food-web system. Some people say that an old city neighborhood where people know each other and communicate regularly is a social system, and that a new apartment block full of strangers is not—not until new relationships arise and a system forms.

A system is more than the sum of its parts. It may exhibit adaptive, dynamic, goal-seeking, self-preserving, and sometimes evolutionary behavior.

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You can see from these examples that there is an integrity or wholeness about a system and an active set of mechanisms to maintain that integrity. Systems can change, adapt, respond to events, seek goals, mend injuries, and attend to their own survival in lifelike ways, although they may contain or consist of nonliving things. Systems can be self-organizing, and often are self-repairing over at least some range of disruptions. They are resilient, and many of them are evolutionary. Out of one system other completely new, never-before-imagined systems can arise.

Look Beyond the Players to the Rules of the Game

> You think that because you understand "one" that you must therefore understand "two" because one and one make two. But you forget that you must also understand "and."
>
> —Sufi teaching story

The elements of a system are often the easiest parts to notice, because many of them are visible, tangible things. The elements that make up a tree are roots, trunk, branches, and leaves. If you look more closely, you see specialized cells: vessels carrying fluids up and down, chloroplasts, and so on. The system called a university is made up of buildings, students, professors, administrators, libraries, books, computers—and I could go on and say what all those things are made up of. Elements do not have to be physical things. Intangibles are also elements of a system. In a university, school pride and academic prowess are two intangibles that can be very important elements of the system. Once you start listing the elements of a system, there is almost no end to the process. You can divide elements into sub-elements and then sub-sub-elements. Pretty soon you lose sight of the system. As the saying goes, you can't see the forest for the trees.

THINK ABOUT THIS

How to know whether you are looking at a system or just a bunch of stuff:

A) Can you identify parts? … and
B) Do the parts affect each other? … and
C) Do the parts together produce an effect that is different from the effect of each part on its own? … and perhaps
D) Does the effect, the behavior over time, persist in a variety of circumstances?

Before going too far in that direction, it's a good idea to stop dissecting out elements and to start looking for the *interconnections*, the relationships that hold the elements together.

The interconnections in the tree system are the physical flows and chemical reactions that govern the tree's metabolic processes—the signals that allow one part to respond to what is happening in another part. For example, as the leaves lose water on a sunny day, a drop in pressure in the water-carrying vessels allows the roots to take in more water. Conversely, if the roots experience dry soil, the loss of water pressure signals the leaves to close their pores, so as not to lose even more precious water.

As the days get shorter in the temperate zones, a deciduous tree puts forth chemical messages that cause nutrients to migrate out of the leaves into the trunk and roots and that weaken the stems, allowing the leaves to fall. There even seem to be messages that cause some trees to make repellent chemicals or harder cell walls if just one part of the plant is attacked by insects. No one understands all the relationships that allow a tree to do what it does. That lack of knowledge is not surprising. It's easier to learn about a system's elements than about its interconnections.

In the university system, interconnections include the standards for admission, the requirements for degrees, the examinations and grades, the budgets and money flows, the gossip, and most important, the communication of knowledge that is, presumably, the purpose of the whole system.

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Many of the interconnections in systems operate through the flow of information. Information holds systems together and plays a great role in determining how they operate.

Some interconnections in systems are actual physical flows, such as the water in the tree's trunk or the students progressing through a university. Many interconnections are flows of information—signals that go to decision points or action points within a system. These kinds of interconnections are often harder to see, but the system reveals them to those who look. Students may use informal information about the probability of getting a good grade to decide what courses to take. A consumer decides what to buy using information about his or her income, savings, credit rating, stock of goods at home, prices, and availability of goods for purchase. Governments need information about kinds and quantities of water pollution before they can create sensible regulations to reduce that pollution. (Note that information about the existence of a problem may be necessary but not sufficient to trigger action—information about resources, incentives, and consequences is necessary too.)

If information-based relationships are hard to see, *functions* or *purposes* are even harder. A system's function or purpose is not necessarily spoken, written, or expressed explicitly, except through the operation of the system. The best way to deduce the system's purpose is to watch for a while to see how the system behaves.

If a frog turns right and catches a fly, and then turns left and catches a fly, and then turns around backward and catches a fly, the purpose of the frog has to do not with turning left or right or backward but with catching flies. If a government proclaims its interest in protecting the environment but allocates little money or effort toward that goal, environmental protection is not, in fact, the government's purpose. Purposes are deduced from behavior, not from rhetoric or stated goals.

A NOTE ON LANGUAGE

The word *function* is generally used for a nonhuman system, the word *purpose* for a human one, but the distinction is not absolute, since so many systems have both human and nonhuman elements.

The function of a thermostat-furnace system is to keep a building at a given temperature. One function of a plant is to bear seeds and create more plants. One purpose of a national economy is, judging from its behavior, to keep growing larger. An important function of almost every system is to ensure its own perpetuation.

System purposes need not be human purposes and are not necessarily those intended by any single actor within the system. In fact, one of the most frustrating aspects of systems is that the purposes of subunits may add up to an overall behavior that no one wants. No one intends to produce a society with rampant drug addiction and crime, but consider the combined purposes and consequent actions of the actors involved:

- desperate people who want quick relief from psychological pain
- farmers, dealers, and bankers who want to earn money
- pushers who are less bound by civil law than are the police who oppose them
- governments that make harmful substances illegal and use police power to interdict them
- wealthy people living in close proximity to poor people
- nonaddicts who are more interested in protecting themselves than in encouraging recovery of addicts

Altogether, these make up a system from which it is extremely difficult to eradicate drug addiction and crime.

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Systems can be nested within systems. Therefore, there can be purposes within purposes. The purpose of a university is to discover and preserve knowledge and pass it on to new generations. Within the university, the purpose of a student may be to get good grades, the purpose of a professor may be to get tenure, the purpose of an administrator may be to balance the budget. Any of those sub-purposes could come into conflict with the overall purpose—the student could cheat, the professor could ignore the students in order to publish papers, the administrator could balance the budget by firing professors. Keeping sub-purposes and overall system purposes in harmony is an essential function of successful systems. I'll get back to this point later when we come to hierarchies.

You can understand the relative importance of a system's elements, interconnections, and purposes by imagining them changed one by one. Changing elements usually has the least effect on the system. If you change all the players on a football team, it is still recognizably a football team. (It may play much better or much worse—particular elements in a system can indeed be important.) A tree changes its cells constantly, its leaves every year or so, but it is still essentially the same tree. Your body replaces most of its cells every few weeks, but it goes on being your body. The university has a constant flow of students and a slower flow of professors and administrators, but it is still a university. In fact it is still the same university, distinct in subtle ways from others, just as General Motors and the U.S. Congress somehow maintain their identities even though all their members change. A system generally goes on being itself, changing only slowly if at all, even with complete substitutions of its elements—as long as its interconnections and purposes remain intact.

The least obvious part of the system, its function or purpose, is often the most crucial determinant of the system's behavior.

If the interconnections change, the system may be greatly altered. It may even become unrecognizable, even though the same players are on the team. Change the rules from those of football to those of basketball, and you've got, as they say, a whole new ball game. If you change the interconnections in the tree—say that instead of taking in carbon dioxide and emitting oxygen, it does the reverse—it would no longer be a tree. (It would be an animal.) If in a university the students graded the professors, or if arguments were won by force instead of reason, the place would need a different name. It might be an interesting organization, but it would not be a university. Changing interconnections in a system can change it dramatically.

Changes in function or purpose also can be drastic. What if you keep the players and the rules but change the purpose—from winning to losing, for example? What if the function of a tree were not to survive and reproduce but to capture all the nutrients in the soil and grow to unlimited size? People have imagined many purposes for a university besides disseminating knowledge—making money, indoctrinating people, winning football games. A change in purpose changes a system profoundly, even if every element and interconnection remains the same.

To ask whether elements, interconnections, or purposes are most important in a system is to ask an unsystemic question. All are essential. All interact. All have their roles. But the least obvious part of the system, its function or purpose, is often the most crucial determinant of the system's behavior. Interconnections are also critically important. Changing relationships usually changes system behavior. The elements, the parts of systems we are most likely to notice, are often (not always) least important in defining the unique characteristics of the system—*unless changing an element also results in changing relationships or purpose*.

Changing just one leader at the top—from a Brezhnev to a Gorbachev, or from a Carter to a Reagan—may or may not turn an entire nation in a new direction, though its land, factories, and hundreds of millions of people remain exactly the same. A leader can make that land and those factories and people play a different game with new rules, or can direct the play toward a new purpose.

And conversely, because land, factories, and people are long-lived, slowly changing, physical elements of the system, there is a limit to the rate at which any leader can turn the direction of a nation.

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Bathtubs 101—Understanding System Behavior over Time

> Information contained in nature … allows us a partial reconstruction of the past.… The development of the meanders in a river, the increasing complexity of the earth's crust … are information-storing devices in the same manner that genetic systems are.… Storing information means increasing the complexity of the mechanism.
>
> —Ramon Margalef[2]

A **stock** is the foundation of any system. Stocks are the elements of the system that you can see, feel, count, or measure at any given time. A system stock is just what it sounds like: a store, a quantity, an accumulation of material or information that has built up over time. It may be the water in a bathtub, a population, the books in a bookstore, the wood in a tree, the money in a bank, your own self-confidence. A stock does not have to be physical. Your reserve of good will toward others or your supply of hope that the world can be better are both stocks.

A stock is the memory of the history of changing flows within the system.

Stocks change over time through the actions of a **flow**. Flows are filling and draining, births and deaths, purchases and sales, growth and decay, deposits and withdrawals, successes and failures. A stock, then, is the present memory of the history of changing flows within the system.



Figure 1. How to read stock-and-flow diagrams. In this book, stocks are shown as boxes, and flows as arrow-headed "pipes" leading into or out of the stocks. The small T on each flow signifies a "faucet;" it can be turned higher or lower, on or off. The "clouds" stand for wherever the flows come from and go to—the sources and sinks that are being ignored for the purposes of the present discussion.

For example, an underground mineral deposit is a stock, out of which comes a flow of ore through mining. The inflow of ore into a mineral deposit is minute in any time period less than eons. So I have chosen to draw (Figure 2) a simplified picture of the system without any inflow. *All* system diagrams and descriptions are simplified versions of the real world.

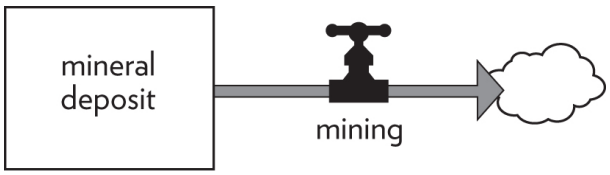


Figure 2. A stock of minerals depleted by mining.

Water in a reservoir behind a dam is a stock, into which flow rain and river water, and out of which flows evaporation from the reservoir's surface as well as the water discharged through the dam.

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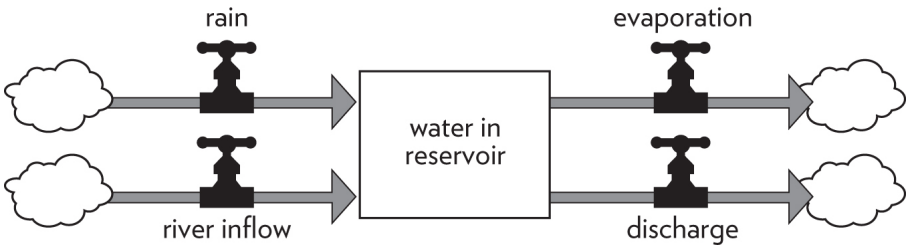


Figure 3. A stock of water in a reservoir with multiple inflows and outflows.

The volume of wood in the living trees in a forest is a stock. Its inflow is the growth of the trees. Its outflows are the natural deaths of trees and the harvest by loggers. The logging harvest flows into another stock, perhaps an inventory of lumber at a mill. Wood flows out of the inventory stock as lumber is sold to customers.

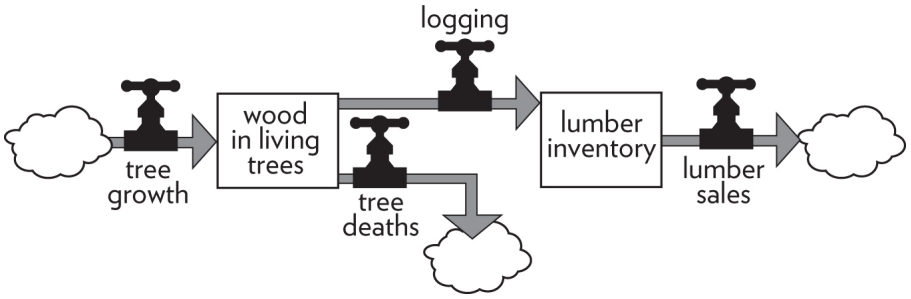


Figure 4. A stock of lumber linked to a stock of trees in a forest.

If you understand the **dynamics** of stocks and flows—their behavior over time—you understand a good deal about the behavior of complex systems. And if you have had much experience with a bathtub, you understand the dynamics of stocks and flows.

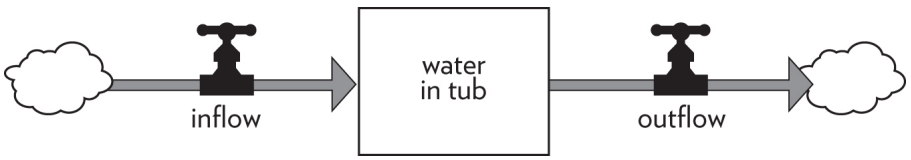


Figure 5. The structure of a bathtub system—one stock with one inflow and one outflow.

Imagine a bathtub filled with water, with its drain plugged up and its faucets turned off—an unchanging, undynamic, boring system. Now mentally pull the plug. The water runs out, of course. The level of water in the tub goes down until the tub is empty.

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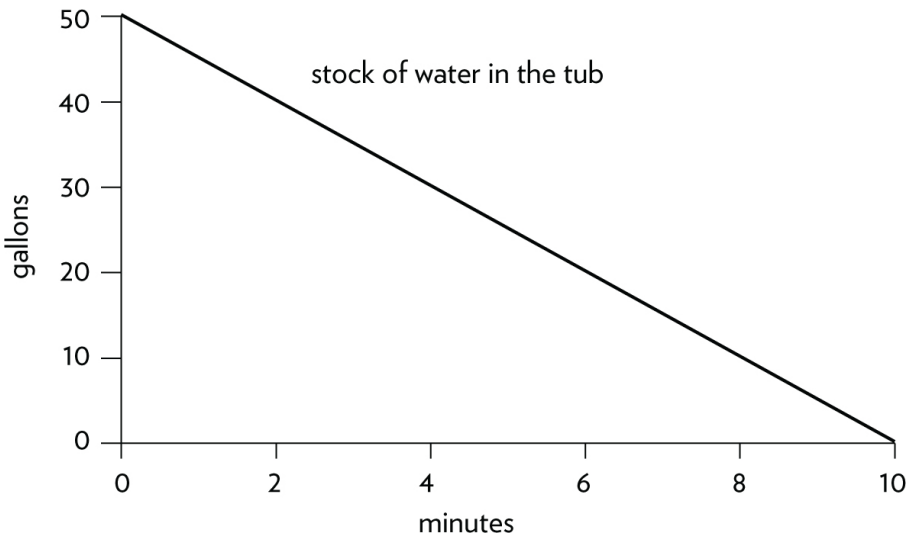


Figure 6. Water level in a tub when the plug is pulled.

A NOTE ON READING GRAPHS OF BEHAVIOR OVER TIME

Systems thinkers use graphs of system behavior to understand trends over time, rather than focusing attention on individual events. We also use behavior-over-time graphs to learn whether the system is approaching a goal or a limit, and if so, how quickly.

The variable on the graph may be a stock or a flow. The pattern—the shape of the variable line—is important, as are the points at which that line changes shape or direction. The precise numbers on the axes are often less important.

The horizontal axis of time allows you to ask questions about what came before, and what might happen next. It can help you focus on the time horizon appropriate to the question or problem you are investigating.

Now imagine starting again with a full tub, and again open the drain, but this time, when the tub is about half empty, turn on the inflow faucet so the rate of water flowing in is just equal to that flowing out. What happens? The amount of water in the tub stays constant at whatever level it had reached when the inflow became equal to the outflow. It is in a state of **dynamic equilibrium**—its level does not change, although water is continuously flowing through it.

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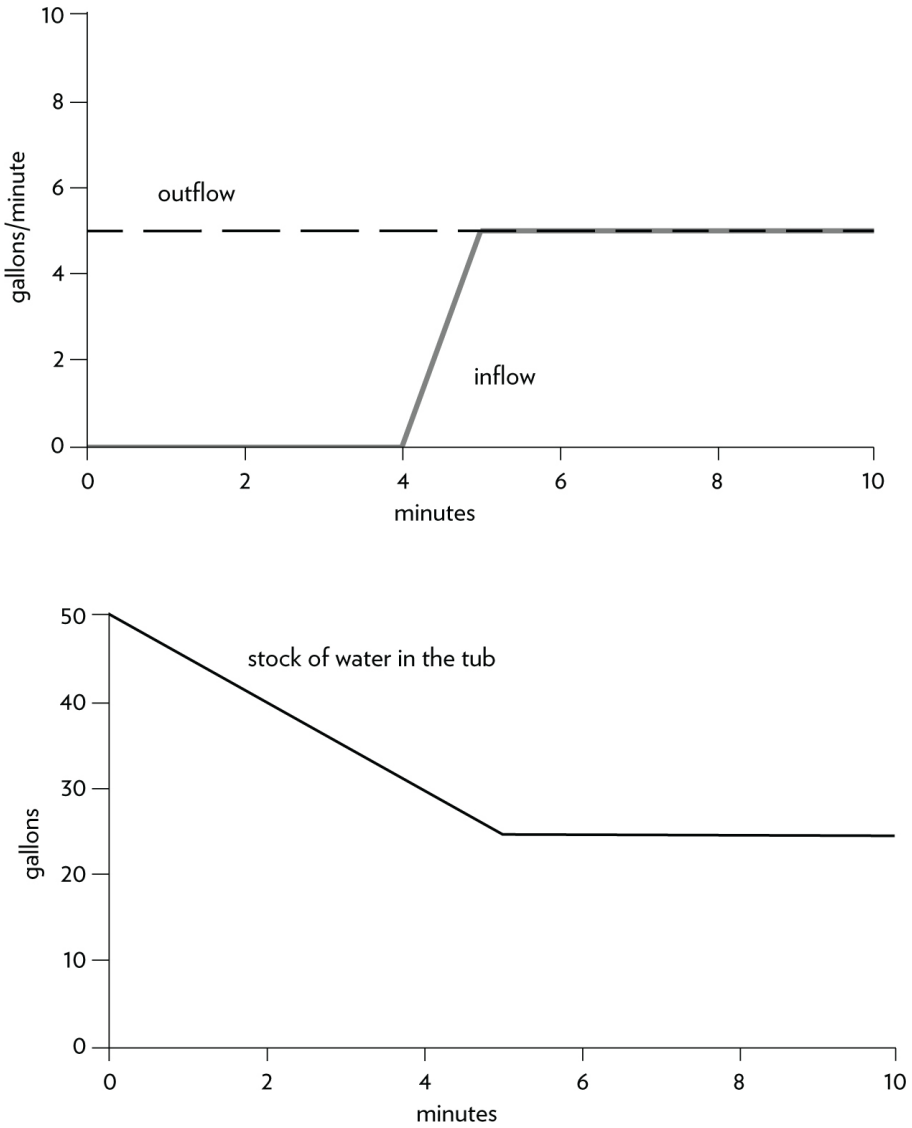


Figure 7. Constant outflow, inflow turned on after 5 minutes, and the resulting changes in the stock of water in the tub.

Imagine turning the inflow on somewhat harder while keeping the outflow constant. The level of water in the tub slowly rises. If you then turn the inflow faucet down again to match the outflow exactly, the water in the tub will stop rising. Turn it down some more, and the water level will fall slowly.

This model of a bathtub is a very simple system with just one stock, one inflow, and one outflow. Over the time period of interest (minutes), I have assumed that evaporation from the tub is insignificant, so I have not included that outflow. All models, whether mental models or mathematical models, are simplifications of the real world. You know all the dynamic possibilities of this bathtub. From it you can deduce several important principles that extend to more complicated systems:

- As long as the sum of all inflows exceeds the sum of all outflows, the level of the stock will rise.
- As long as the sum of all outflows exceeds the sum of all inflows, the level of the stock will fall.
- If the sum of all outflows equals the sum of all inflows, the stock level will not change; it will be held in dynamic equilibrium at whatever level it happened to be when the two sets of flows became equal.

The human mind seems to focus more easily on stocks than on flows. On top of that, when we do focus on flows, we tend to focus on *inflows* more easily than on outflows. Therefore, we sometimes miss seeing that we can fill a bathtub not only by increasing the inflow rate, but also by decreasing the outflow rate. Everyone understands that you can prolong the life of an oil-based economy by discovering new oil

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deposits. It seems to be harder to understand that the same result can be achieved by burning less oil. A breakthrough in energy efficiency is equivalent, in its effect on the stock of available oil, to the discovery of a new oil field—although different people profit from it.

A stock can be increased by decreasing its outflow rate as well as by increasing its inflow rate. There's more than one way to fill a bathtub!

Similarly, a company can build up a larger workforce by more hiring, or it can do the same thing by reducing the rates of quitting and firing. These two strategies may have very different costs. The wealth of a nation can be boosted by investment to build up a larger stock of factories and machines. It also can be boosted, often more cheaply, by decreasing the rate at which factories and machines wear out, break down, or are discarded.

You can adjust the drain or faucet of a bathtub—the flows—abruptly, but it is much more difficult to change the level of water—the stock—quickly. Water can't run out the drain instantly, even if you open the drain all the way. The tub can't fill up immediately, even with the inflow faucet on full blast. *A stock takes time to change, because flows take time to flow*. That's a vital point, a key to understanding why systems behave as they do. Stocks usually change slowly. They can act as delays, lags, buffers, ballast, and sources of momentum in a system. Stocks, especially large ones, respond to change, even sudden change, only by gradual filling or emptying.

Stocks generally change slowly, even when the flows into or out of them change suddenly. Therefore, **stocks act as delays or buffers or shock absorbers in systems**.

People often underestimate the inherent momentum of a stock. It takes a long time for populations to grow or stop growing, for wood to accumulate in a forest, for a reservoir to fill up, for a mine to be depleted. An economy cannot build up a large stock of functioning factories and highways and electric plants overnight, even if a lot of money is available. Once an economy has a lot of oil-burning furnaces and automobile engines, it cannot change quickly to furnaces and engines that burn a different fuel, even if the price of oil suddenly changes. It has taken decades to accumulate the stratospheric pollutants that destroy the earth's ozone layer; it will take decades for those pollutants to be removed.

Changes in stocks set the pace of the dynamics of systems. Industrialization cannot proceed faster than the rate at which factories and machines can be constructed and the rate at which human beings can be educated to run and maintain them. Forests can't grow overnight. Once contaminants have accumulated in groundwater, they can be washed out only at the rate of groundwater turnover, which may take decades or even centuries.

The time lags that come from slowly changing stocks can cause problems in systems, but they also can be sources of stability. Soil that has accumulated over centuries rarely erodes all at once. A population that has learned many skills doesn't forget them immediately. You can pump groundwater faster than the rate it recharges for a long time before the aquifer is drawn down far enough to be damaged. The time lags imposed by stocks allow room to maneuver, to experiment, and to revise policies that aren't working.

If you have a sense of the rates of change of stocks, you don't expect things to happen faster than they can happen. You don't give up too soon. You can use the opportunities presented by a system's momentum to guide it toward a good outcome—much as a judo expert uses the momentum of an opponent to achieve his or her own goals.

There is one more important principle about the role of stocks in systems, a principle that will lead us directly to the concept of feedback. The presence of stocks allows inflows and outflows to be independent of each other and temporarily out of balance with each other.

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Stocks allow inflows and outflows to be decoupled and to be independent and temporarily out of balance with each other.

It would be hard to run an oil company if gasoline had to be produced at the refinery at exactly the rate the cars were burning it. It isn't feasible to harvest a forest at the precise rate at which the trees are growing. Gasoline in storage tanks and wood in the forest are both stocks that permit life to proceed with some certainty, continuity, and predictability, even though flows vary in the short term.

Human beings have invented hundreds of stock-maintaining mechanisms to make inflows and outflows independent and stable. Reservoirs enable residents and farmers downriver to live without constantly adjusting their lives and work to a river's varying flow, especially its droughts and floods. Banks enable you temporarily to earn money at a rate different from how you spend. Inventories of products along a chain from distributors to wholesalers to retailers allow production to proceed smoothly although customer demand varies, and allow customer demand to be filled even though production rates vary.

Most individual and institutional decisions are designed to regulate the levels in stocks. If inventories rise too high, then prices are cut or advertising budgets are increased, so that sales will go up and inventories will fall. If the stock of food in your kitchen gets low, you go to the store. As the stock of growing grain rises or fails to rise in the fields, farmers decide whether to apply water or pesticide, grain companies decide how many barges to book for the harvest, speculators bid on future values of the harvest, cattle growers build up or cut down their herds. Water levels in reservoirs cause all sorts of corrective actions if they rise too high or fall too low. The same can be said for the stock of money in your wallet, the oil reserves owned by an oil company, the pile of woodchips feeding a paper mill, and the concentration of pollutants in a lake.

People monitor stocks constantly and make decisions and take actions designed to raise or lower stocks or to keep them within acceptable ranges. Those decisions add up to the ebbs and flows, successes and problems, of all sorts of systems. Systems thinkers see the world as a collection of stocks along with the mechanisms for regulating the levels in the stocks by manipulating flows.

That means system thinkers see the world as a collection of "feedback processes."

How the System Runs Itself—Feedback

> Systems of information-feedback control are fundamental to all life and human endeavor, from the slow pace of biological evolution to the launching of the latest space satellite.… Everything we do as individuals, as an industry, or as a society is done in the context of an information-feedback system.
>
> —Jay W. Forrester[3]

When a stock grows by leaps and bounds or declines swiftly or is held within a certain range no matter what else is going on around it, it is likely that there is a control mechanism at work. In other words, if you see a behavior that persists over time, there is likely a mechanism creating that consistent behavior. That mechanism operates through a **feedback loop**. It is the consistent behavior pattern over a long period of time that is the first hint of the existence of a feedback loop.

A feedback loop is formed when changes in a stock affect the flows into or out of that same stock. A feedback loop can be quite simple and direct. Think of an interest-bearing savings account in a bank. The total amount of money in the account (the stock) affects how much money comes into the account as interest. That is because the bank has a rule that the account earns a certain percent interest each year. The total dollars of interest paid into the account each year (the flow in) is not a fixed amount, but varies with the size of the total in the account.

You experience another fairly direct kind of feedback loop when you get your bank statement for your checking account each month. As your level of available cash in the checking account (a stock) goes down, you may decide to work more hours and earn more money. The money entering your bank account is a flow that you can adjust in order to increase your stock of cash to a more desirable level. If your bank account then grows very large, you may feel free to work less (decreasing the inflow). This kind of feedback loop is

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keeping your level of cash available within a range that is acceptable to you. You can see that adjusting your earnings is not the only feedback loop that works on your stock of cash. You also may be able to adjust the outflow of money from your account, for example. You can imagine an outflow-adjusting feedback loop for spending.

Feedback loops can cause stocks to maintain their level within a range or grow or decline. In any case, the flows into or out of the stock are adjusted because of changes in the size of the stock itself. Whoever or whatever is monitoring the stock's level begins a corrective process, adjusting rates of inflow or outflow (or both) and so changing the stock's level. The stock level feeds back through a chain of signals and actions to control itself.

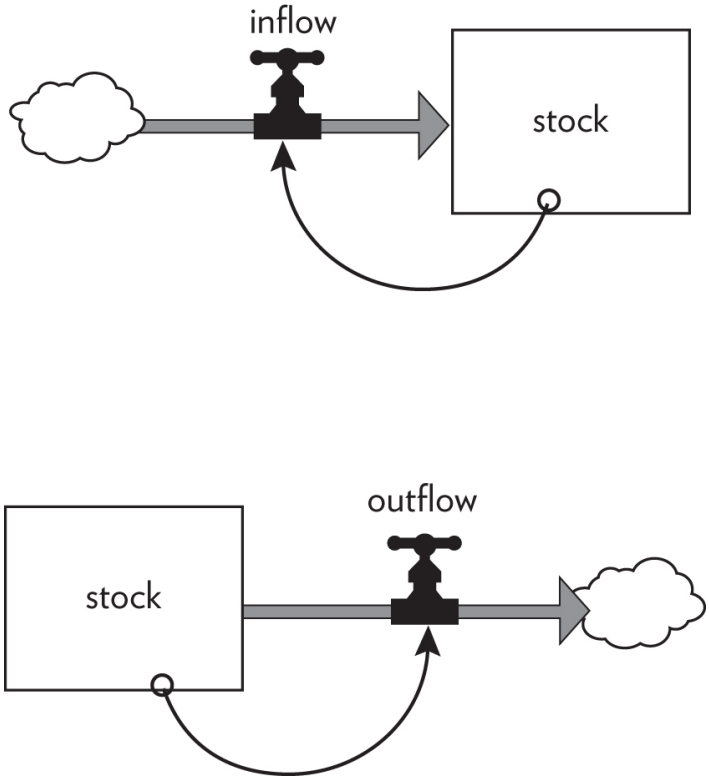


Figure 8. How to read a stock-and-flow diagram with feedback loops. Each diagram distinguishes the stock, the flow that changes the stock, and the information link (shown as a thin, curved arrow) that directs the action. It emphasizes that action or change always proceeds through adjusting flows.

Not all systems have feedback loops. Some systems are relatively simple open-ended chains of stocks and flows. The chain may be affected by outside factors, but the levels of the chain's stocks don't affect its flows. However, those systems that contain feedback loops are common and may be quite elegant or rather surprising, as we shall see.

A feedback loop is a closed chain of causal connections from a stock, through a set of decisions or rules or physical laws or actions that are dependent on the level of the stock, and back again through a flow to change the stock.

Stabilizing Loops—Balancing Feedback

One common kind of feedback loop stabilizes the stock level, as in the checking-account example. The stock level may not remain completely fixed, but it does stay within an acceptable range. What follows are

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some more stabilizing feedback loops that may be familiar to you. These examples start to detail some of the steps within a feedback loop.

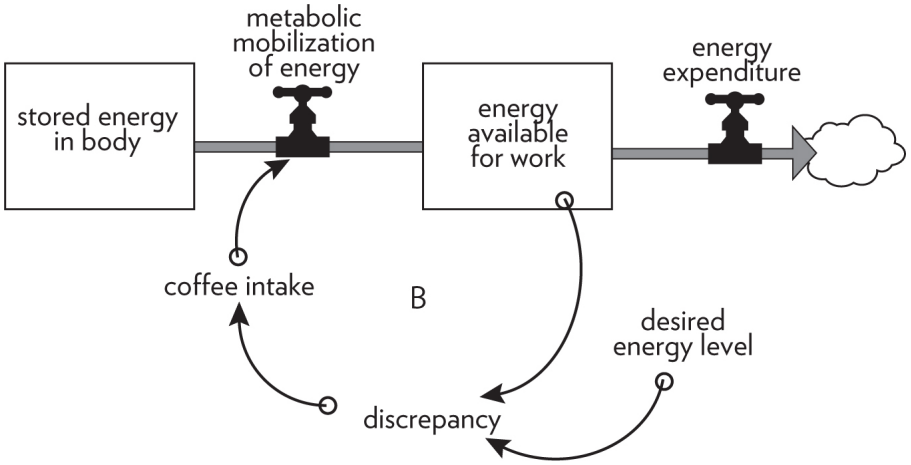


Figure 9. Energy level of a coffee drinker.

If you're a coffee drinker, when you feel your energy level run low, you may grab a cup of hot black stuff to perk you up again. You, as the coffee drinker, hold in your mind a desired stock level (energy for work). The purpose of this caffeine-delivery system is to keep your actual stock level near or at your desired level. (You may have other purposes for drinking coffee as well: enjoying the flavor or engaging in a social activity.) It is the gap, the discrepancy, between your actual and desired levels of energy for work that drives your decisions to adjust your daily caffeine intake.

Notice that the labels in Figure 9, like all the diagram labels in this book, are direction-free. The label says "stored energy in body" not "*low* energy level," "coffee intake" not "*more* coffee." That's because feedback loops often can operate in two directions. In this case, the feedback loop can correct an oversupply as well as an undersupply. If you drink too much coffee and find yourself bouncing around with extra energy, you'll lay off the caffeine for a while. High energy creates a discrepancy that says "too much," which then causes you to reduce your coffee intake until your energy level settles down. The diagram is intended to show that the loop works to drive the stock of energy in either direction.

I could have shown the inflow of energy coming from a cloud, but instead I made the system diagram slightly more complicated. *Remember—all system diagrams are simplifications of the real world*. We each choose how much complexity to look at. In this example, I drew another stock—the stored energy in the body that can be activated by the caffeine. I did that to indicate that there is more to the system than one simple loop. As every coffee drinker knows, caffeine is only a short-term stimulant. It lets you run your motor faster, but it doesn't refill your personal fuel tank. Eventually the caffeine high wears off, leaving the body more energy-deficient than it was before. That drop could reactivate the feedback loop and generate another trip to the coffee pot. (See the discussion of addiction later in this book.) Or it could activate some longer-term and healthier feedback responses: Eat some food, take a walk, get some sleep.

This kind of stabilizing, goal-seeking, regulating loop is called a **balancing feedback loop**, so I put a B inside the loop in the diagram. Balancing feedback loops are *goal-seeking* or *stability-seeking*. Each tries to keep a stock at a given value or within a range of values. A balancing feedback loop opposes whatever direction of change is imposed on the system. If you push a stock too far up, a balancing loop will try to pull it back down. If you shove it too far down, a balancing loop will try to bring it back up.

Here's another balancing feedback loop that involves coffee, but one that works through physical law rather than human decision. A hot cup of coffee will gradually cool down to room temperature. Its rate of cooling depends on the difference between the temperature of the coffee and the temperature of the room. The greater the difference, the faster the coffee will cool. The loop works the other way too—if you make iced coffee on a hot day, it will warm up until it has the same temperature as the room. The function of this system is to bring the discrepancy between coffee's temperature and room's temperature to zero, no matter what the direction of the discrepancy.

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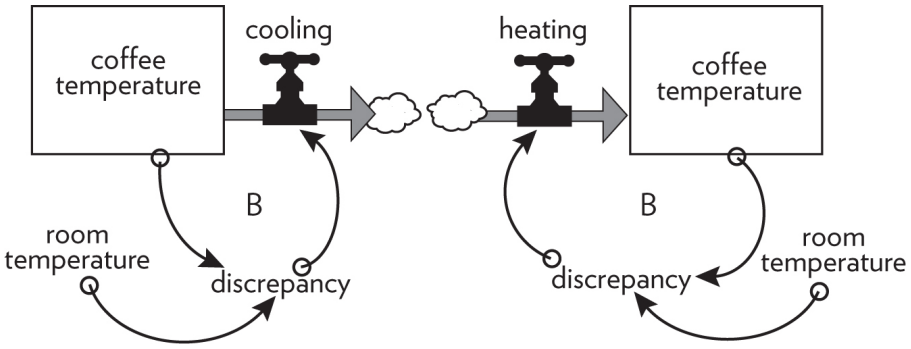


Figure 10. A cup of coffee cooling (*left*) or warming (*right*).

Starting with coffee at different temperatures, from just below boiling to just above freezing, the graphs in Figure 11 show what will happen to the temperature over time (if you don't drink the coffee). You can see here the "homing" behavior of a balancing feedback loop. Whatever the initial value of the system stock (coffee temperature in this case), whether it is above or below the "goal" (room temperature), the feedback loop brings it toward the goal. The change is faster at first, and then slower, as the discrepancy between the stock and the goal decreases.

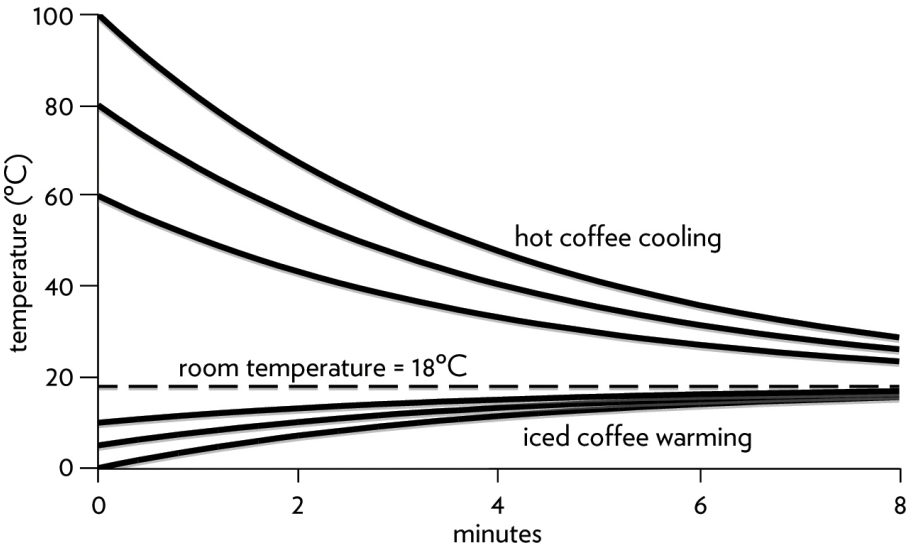


Figure 11. Coffee temperature as it approaches a room temperature of 18°C.

This behavior pattern—gradual approach to a system-defined goal— also can be seen when a radioactive element decays, when a missile finds its target, when an asset depreciates, when a reservoir is brought up or down to its desired level, when your body adjusts its blood-sugar concentration, when you pull your car to a stop at a stoplight. You can think of many more examples. The world is full of goal-seeking feedback loops.

Balancing feedback loops are equilibrating or goal-seeking structures in systems and are both sources of stability and sources of resistance to change.

The presence of a feedback mechanism doesn't necessarily mean that the mechanism works *well*. The feedback mechanism may not be strong enough to bring the stock to the desired level. Feedbacks—the interconnections, the information part of the system—can fail for many reasons. Information can arrive too late or at the wrong place. It can be unclear or incomplete or hard to interpret. The action it triggers may be too weak or delayed or resource-constrained or simply ineffective. The goal of the feedback loop may never

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be reached by the actual stock. But in the simple example of a cup of coffee, the drink eventually will reach room temperature.

Runaway Loops—Reinforcing Feedback

> I'd need rest to refresh my brain, and to get rest it's necessary to travel, and to travel one must have money, and in order to get money you have to work.… I am in a vicious circle … from which it is impossible to escape.
>
> —Honoré Balzac,[4] 19th century novelist and playwright

> Here we meet a very important feature. It would seem as if this were circular reasoning; profits fell because investment fell, and investment fell because profits fell.
>
> —Jan Tinbergen,[5] economist

The second kind of feedback loop is amplifying, reinforcing, self-multiplying, snowballing—a vicious or virtuous circle that can cause healthy growth or runaway destruction. It is called a **reinforcing feedback loop**, and will be noted with an R in the diagrams. It generates more input to a stock the more that is already there (and less input the less that is already there). A reinforcing feedback loop enhances whatever direction of change is imposed on it.

For example:

- When we were kids, the more my brother pushed me, the more I pushed him back, so the more he pushed me back, so the more I pushed him back.
- The more prices go up, the more wages have to go up if people are to maintain their standards of living. The more wages go up, the more prices have to go up to maintain profits. This means that wages have to go up again, so prices go up again.
- The more rabbits there are, the more rabbit parents there are to make baby rabbits. The more baby rabbits there are, the more grow up to become rabbit parents, to have even more baby rabbits.
- The more soil is eroded from the land, the less plants are able to grow, so the fewer roots there are to hold the soil, so the more soil is eroded, so less plants can grow.
- The more I practice piano, the more pleasure I get from the sound, and so the more I play the piano, which gives me more practice.

Reinforcing loops are found wherever a system element has the ability to reproduce itself or to grow as a constant fraction of itself. Those elements include populations and economies. Remember the example of the interest-bearing bank account? The more money you have in the bank, the more interest you earn, which is added to the money already in the bank, where it earns even more interest.

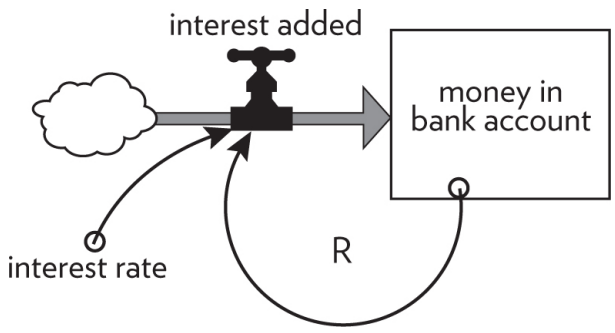


Figure 12. Interest-bearing bank account.

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